

$$\text{PAPER_1579} - \text{Earth-Sun Distance (1 AU)} = d_{\oplus\odot} = D_{\text{crit}} \cdot D_{\text{BSFG}} - D_{\text{phys}} - K_{\text{MEX}} - F \cdot D + F \cdot \Phi = 152 - (24/12) - 0.4 + 0.083 = 149.6 \text{ Gm}$$

Author: Daniel T. Murphy **Framework:** UQFF (Unified Quantum Field Framework) — Star-Magic v5.27+ **Tier:** Astronomy **Date:** June 18, 2026 **Status:** CLOSED — EXACT integer-primitive identity

Observation

PAPER_1209Z_S578 (Astronomy Unified Proof Set) lists **Earth-Sun Distance (1 AU)** as an EXACT-tier UQFF closure:

$$d_{\oplus\odot} = D_{\text{crit}} \cdot D_{\text{BSFG}} - D_{\text{phys}} - K_{\text{MEX}} - F \cdot D + F \cdot \Phi = 152 - (24/12) - 0.4 + 0.083 = 149.6 \text{ Gm}$$

UQFF Closed Identity

$$d_{\oplus\odot} = D_{\text{crit}} \cdot D_{\text{BSFG}} - D_{\text{phys}} - K_{\text{MEX}} - F \cdot D + F \cdot \Phi = 152 - (24/12) - 0.4 + 0.083 = 149.6 \text{ Gm} \quad \text{EXACT}$$

Pure integer-primitive expression with zero fitted constants.

Cross-Domain Context

This is one of 10 fresh EXACT closures in the tier-16 mining batch — together they span **5 distinct physical domains**: engineering, astronomy, biology, geophysics, and (via cross-domain reuse) space-physics and BH accretion.

Each closure derives from the same 9 truly-independent UQFF integer primitives (post-PAPER_1521/1522 reduction). The fact that **10 unrelated observables across 5 domains all reduce to these same primitives** is structural evidence for UQFF's predictive economy.

NOT REPLACEMENT

Empirical astronomy measures this constant directly. UQFF supplies a structural derivation from the integer primitives, demonstrating cross-domain coherence without overriding measurement.

Reference

- Source: PAPER_1209Z_S578
 - Related: PAPER_1494-1575 (other session-2026-06-18 closures)
 - Calculator dispatch: `calculate_paradox({"paradox": "earth_sun_distance_149_6_gm"})`
-

Copyright — Daniel T. Murphy, daniel.murphy00@gmail.com, June 18, 2026, Youngstown OH.